PRACTICAL-06[OS]

INPUT:

```c
include <stdio.h>
include <stdlib.h>
include <string.h>
include <fcntl.h>
include <unistd.h>
include <sys/stat.h>

define SERVER_FIFO "server_fifo"
define CLIENT_FIFO "client_fifo"

nt main()

   char message[100];
   char response[100];

   // Create named pipes
   mkfifo(SERVER_FIFO, 0666);
   mkfifo(CLIENT_FIFO, 0666);

   printf("Server started...\n");
   printf("Waiting for client message...\n");

   // Open server FIFO for reading
   int server_fd = open(SERVER_FIFO, O_RDONLY);

   // Read message from client
   read(server_fd, message, sizeof(message));

   printf("Client says: %s\n", message);

   close(server_fd);

   // Prepare response
   sprintf(response, "Server received your message: %s", message);

   // Open client FIFO for writing
   int client_fd = open(CLIENT_FIFO, O_WRONLY);
```

```
^G Help        ^O Write Out    ^F Where Is     ^K Cut          ^T Execute      ^C Location     M-U Undo        M-A Set Mark    M-] To Bracket  M-B Previous
^X Exit        ^R Read File    ^\ Replace      ^U Paste        ^J Justify      ^/ Go To Line   M-E Redo        M-6 Copy        ^B Where Was    M-F Next
```

```c
{
    char message[100];
    char response[100];

    // Create named pipes
    mkfifo(SERVER_FIFO, 0666);
    mkfifo(CLIENT_FIFO, 0666);

    printf("Server started...\n");
    printf("Waiting for client message...\n");

    // Open server FIFO for reading
    int server_fd = open(SERVER_FIFO, O_RDONLY);

    // Read message from client
    read(server_fd, message, sizeof(message));

    printf("Client says: %s\n", message);

    close(server_fd);

    // Prepare response
    sprintf(response, "Server received your message: %s", message);

    // Open client FIFO for writing
    int client_fd = open(CLIENT_FIFO, O_WRONLY);

    // Send response
    write(client_fd, response, strlen(response) + 1);

    close(client_fd);

    printf("Response sent to client.\n");

    return 0;
}
```

```
^G Help        ^O Write Out    ^F Where Is     ^K Cut          ^T Execute      ^C Location     M-U Undo        M-A Set Mark    M-] To Bracket  M-B Previous
^X Exit        ^R Read File    ^\ Replace      ^U Paste        ^J Justify      ^/ Go To Line   M-E Redo        M-6 Copy        ^B Where Was    M-F Next
```



```c
#include <unistd.h>

#define SERVER_FIFO "server_fifo"
#define CLIENT_FIFO "client_fifo"

int main()
{
    char message[100];
    char response[100];

    printf("Enter message for server: ");
    fgets(message, sizeof(message), stdin);

    // Remove newline
    message[strcspn(message, "\n")] = '\0';

    // Open server FIFO for writing
    int server_fd = open(SERVER_FIFO, O_WRONLY);

    // Send message
    write(server_fd, message, strlen(message) + 1);

    close(server_fd);

    // Open client FIFO for reading
    int client_fd = open(CLIENT_FIFO, O_RDONLY);

    // Read server response
    read(client_fd, response, sizeof(response));

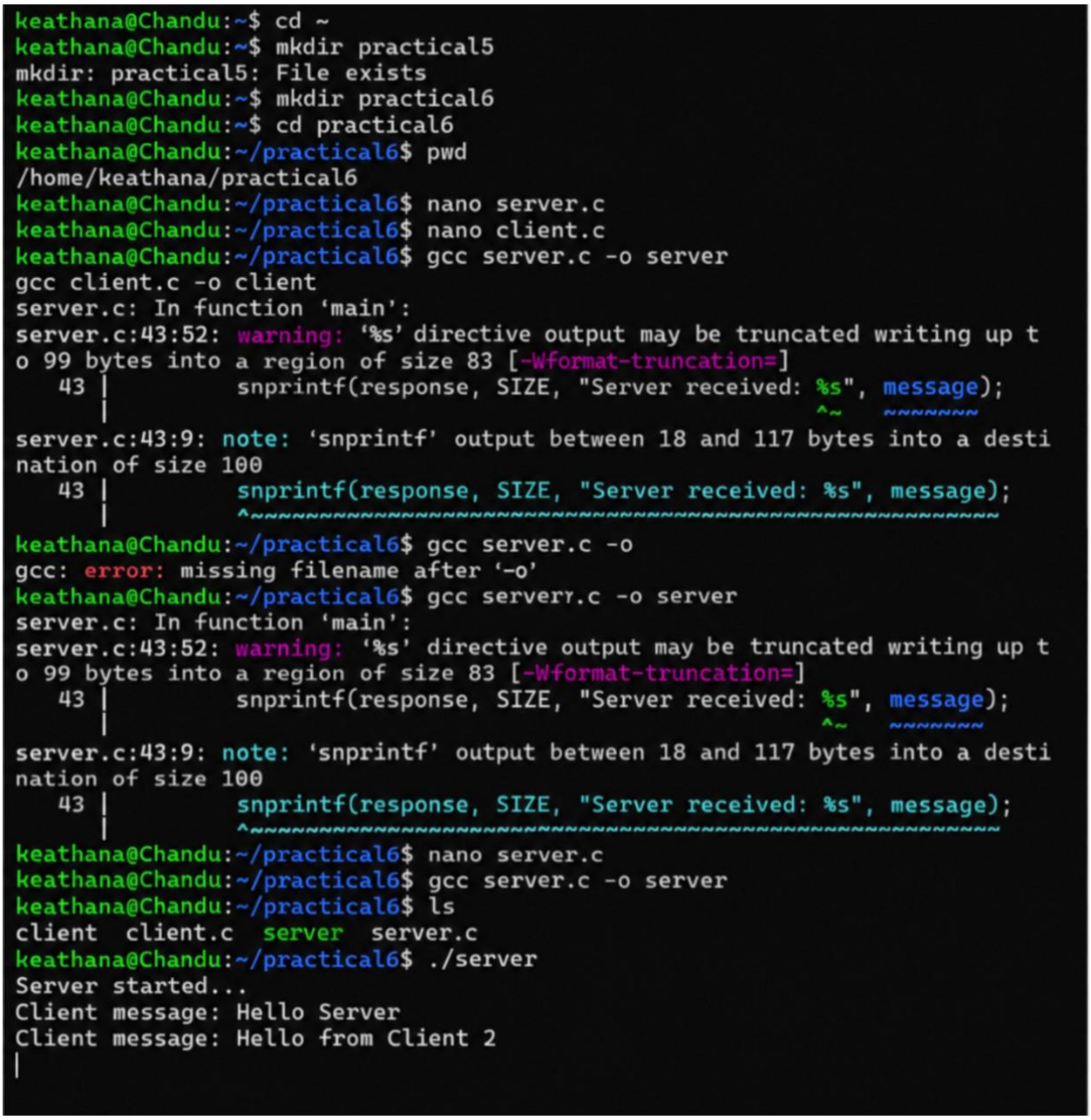
    printf("Server response: %s\n", response);

    close(client_fd);

    return 0;
}
```

```
^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T Execute       ^C Location      M-U Undo         M-A Set Mark     M-] To Bracket   M-B Previous
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J Justify       ^/ Go To Line    M-E Redo         M-G Copy         ^B Where Was     M-F Next
```

OUTPUT:


keathana@Chandu:~$ cd ~
keathana@Chandu:~$ mkdir practical5
mkdir: practical5: File exists
keathana@Chandu:~$ mkdir practical6
keathana@Chandu:~$ cd practical6
keathana@Chandu:~/practical6$ pwd
/home/keathana/practical6
keathana@Chandu:~/practical6$ nano server.c
keathana@Chandu:~/practical6$ nano client.c
keathana@Chandu:~/practical6$ gcc server.c -o server
gcc client.c -o client
server.c: In function 'main':
server.c:43:52: warning: '%s' directive output may be truncated writing up t
o 99 bytes into a region of size 83 [-Wformat-truncation=]
43 | snprintf(response, SIZE, "Server received: %s", message);
server.c:43:9: note: 'snprintf' output between 18 and 117 bytes into a desti
nation of size 100
43 | snprintf(response, SIZE, "Server received: %s", message);
keathana@Chandu:~/practical6$ gcc server.c -o
gcc: error: missing filename after '-o'
keathana@Chandu:~/practical6$ gcc serverr.c -o server
server.c: In function 'main':
server.c:43:52: warning: '%s' directive output may be truncated writing up t
o 99 bytes into a region of size 83 [-Wformat-truncation=]
43 | snprintf(response, SIZE, "Server received: %s", message);
server.c:43:9: note: 'snprintf' output between 18 and 117 bytes into a desti
nation of size 100
43 | snprintf(response, SIZE, "Server received: %s", message);
keathana@Chandu:~/practical6$ nano server.c
keathana@Chandu:~/practical6$ gcc server.c -o server
keathana@Chandu:~/practical6$ ls
client client.c server server.c
keathana@Chandu:~/practical6$ ./server
Server started...
Client message: Hello Server
Client message: Hello from Client 2

```
keathana@Chandu:~$ cd ~/practical6
keathana@Chandu:~/practical6$ ./client
Enter message: Hello Server
Server response: Server received: Hello Server
keathana@Chandu:~/practical6$
```

```
keathana@Chandu:~$ cd ~/practical6
keathana@Chandu:~/practical6$ ./client
Enter message: Hello from Client 2
Server response: Server received: Hello from Client 2
keathana@Chandu:~/practical6$
```